

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE:** CSA13

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:25 Duration : 1 Hour

Analytical Problem

Code of Conduct:

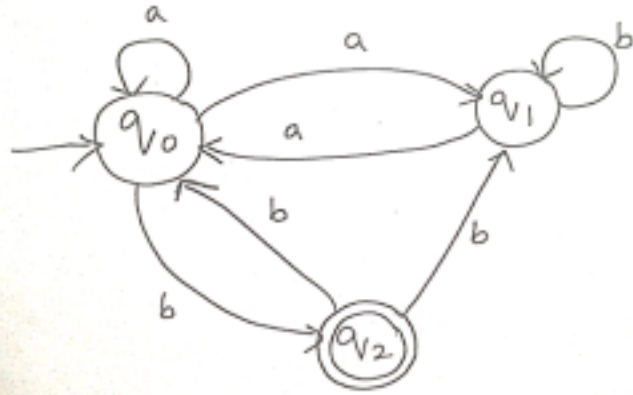
I Kadava Yogi (192372053) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action. Signature of the student:K.Yogi

SIMATS ENGINEERING ASSESSMENT TOOL

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L

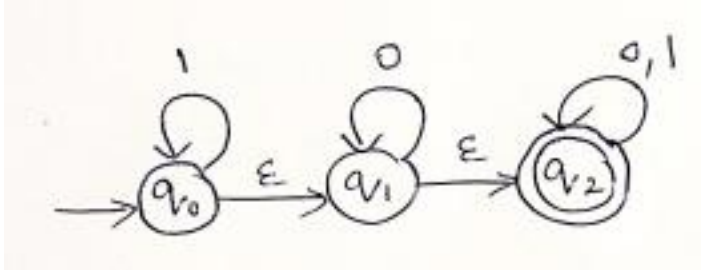


2. Let L be the language of words of length atleast 2 whose last two letters are different from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



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ASSESSMENT TOOL**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Construct the Equivalent DFA for the Given NFA

NFA Transitions (from the figure)

State	a	b
→q0	{q0,q1}	{q2}
q1	{q0}	{q1,q2}
*q2	∅	{q0}

Subset Construction

DFA States

- **A = {q0}** (Start)
- **B = {q0,q1}**
- **C = {q2}**
- **D = {q0,q1,q2}**
- **E = ∅** (Dead)

Final states are those containing **q2**:

- **C**
- **D**

DFA Transition Table

DFA State	a	b
→A={q0}	B	C
B={q0,q1}	B	D
*C={q2}	E	A
*D={q0,q1,q2}	B	D
E=∅	E	E

2. NFA and DFA for Strings of Length ≥ 2 whose Last Two Symbols are Different

Language:

$$L = (a + b)^*(ab + ba)$$

NFA

States

- q0 (Start)
- q1 (seen a)
- q2 (seen b)
- q3 (Final)

Transitions

State	a	b
→q0	q0,q1	q0,q2
q1	-	q3
q2	q3	-
*q3	-	-

Equivalent DFA

States

- q_0 (Start)
- q_1 (Last=a)
- q_2 (Last=b)
- q_3 (Ends with ab) ✓
- q_4 (Ends with ba) ✓

State a b

→ q_0 q_1 q_2

q_1 q_1 q_3

q_2 q_4 q_2

* q_3 q_4 q_2

* q_4 q_1 q_3

Final States = **q_3, q_4**

3. NFA for Binary Strings Whose Last Two Digits are Same

Language

$$(0 + 1)^*(00 + 11)$$

Formal Definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- Start state = q_0
- Final states = $\{q_2, q_4\}$

Transition Table

State 0 1

→ q_0 q_0, q_1 q_0, q_3

q_1 q_2 -

* q_2 - -

q_3 - q_4

* q_4 - -

Acceptance of 01011

q_0

--0--> q_0

--1--> q_0

--0--> q_0

--1--> q_3

--1--> q_4 (Accepted)

Hence,

01011 is accepted.

4. DFA for Strings Containing 001 as a Substring

States

- q_0 (Start)
- q_1 (Seen 0)
- q_2 (Seen 00)
- q_3 (Seen 001) ✓

Transition Table

State 0 1

$\rightarrow q_0$ q_1 q_0

q_1 q_2 q_0

q_2 q_2 q_3

$*q_3$ q_3 q_3

State Diagram (Text)

$\rightarrow q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_2 \xrightarrow{1} *q_3$

^			
1	1	0	0,1
+-----+	+-----+	+-----+	+-----+

5. ϵ -Closure and Equivalent DFA

ϵ -NFA

Transitions

- $q_0 \xrightarrow{\epsilon} q_1$
- $q_1 \xrightarrow{\epsilon} q_2$
- q_0 loop on 1
- q_1 loop on 0
- q_2 loop on 0,1

ϵ -Closures

State ϵ -Closure

q_0 { q_0, q_1, q_2 }

q_1 { q_1, q_2 }

q_2 { q_2 }

Equivalent DFA

Let

- $A = \{q_0, q_1, q_2\}$ (Start)
- $B = \{q_1, q_2\}$
- $C = \{q_2\}$

Since each contains q_2 , all are final states.

Transition Table

State 0 1

$\rightarrow *A$ B A

State 0 1

*B B C

*C C C

Final States

A, B, C